

Assessing and Mapping Urban Tree Canopy Coverage Using GIS for Sustainable Green Infrastructure

A Case Study of New York City



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Date of Submission:

May 6th, 2025

Declaration:

We hereby declare that the work presented in this report titled "Assessing and Mapping Urban Tree Canopy Coverage Using GIS for Sustainable Green Infrastructure – A Case Study of New York City" is original and carried out by us as part of our course requirement. All the data sources and references used have been acknowledged appropriately.

Acknowledgements:

We would like to express our sincere gratitude to our course instructor Jackson LaSarlo **for** their valuable guidance and continuous support throughout this project. We also acknowledge the use of free and open-source data from USGS, NYC Open Data, and the use of ArcGIS/QGIS software provided by the university.

Abstract:

Urban tree canopy (UTC) is a critical component of sustainable green infrastructure, offering a range of environmental, social, and health benefits in dense metropolitan settings. This project assesses and maps the urban tree canopy of New York City for the year 2021 using Geographic Information Systems (GIS), remote sensing, and spatial statistical methods. The analysis integrates NDVI (Normalized Difference Vegetation Index) and Land Surface Temperature (LST) data derived from Landsat 8 imagery, complemented by high-resolution tree canopy raster datasets.

The methodology includes processing satellite data, visualizing vegetation and temperature distributions, and applying Ordinary Least Squares (OLS), Spatial Lag Model (SLM), and Spatial Error Model (SEM) regressions to investigate the statistical relationship between canopy coverage and urban heat. Results show that approximately 12.63% of the study area is classified as tree canopy. However, regression analysis for 2021 reveals no statistically significant relationship between canopy density and LST, although spatial autocorrelation is strong across the city.

These findings highlight the complexity of urban heat patterns and suggest that factors beyond tree canopy, such as land use, building density, and impervious surfaces, also influence LST. Despite this, the ecological and public health value of urban trees supports the continued prioritization of canopy expansion in city planning. This study demonstrates the utility of spatial data integration and modeling to inform green infrastructure strategies and urban environmental resilience efforts.

1. INTRODUCTION

1.1 What is Urban Tree Canopy (UTC)?

Urban Tree Canopy (UTC) refers to the layer of tree leaves, branches, and stems that cover the ground when viewed from above. This canopy layer plays a critical ecological role by providing shade, sequestering carbon, intercepting stormwater, enhancing air quality, and supporting urban biodiversity. In metropolitan settings like New York City, where built infrastructure dominates the landscape, UTC serves as a crucial natural asset that offers both environmental and societal benefits. Tree canopy can exist in various forms, trees, park forests, backyard vegetation, and scattered tree clusters—and its extent is often measured as a percentage of land area covered by tree foliage. As urban populations grow and climate-related challenges intensify, assessing the extent, distribution, and functionality of UTC becomes essential for sustainable urban planning and policymaking.



A visual summary showcasing how urban tree canopy mitigate climate change, reduces energy use, and improves public health.

Tree-lined urban streets provide shade, comfort, and encourage walkability in densely populated areas.



1.2 Importance of UTC in Urban Areas and Sustainable Green Infrastructure

Urban environments around the world are increasingly grappling with the challenges posed by rapid development, climate change, and socio-economic disparities. Among the many strategies to improve urban sustainability and livability, the conservation and expansion of UTC is considered one of the most impactful and cost-effective. In cities such as New York, which is characterized by dense built environments and unevenly distributed green spaces, tree canopy plays a crucial role in enhancing resilience and improving the quality of life for residents.

The benefits of UTC are multifaceted. Environmentally, tree canopies moderate temperatures through shade and evapotranspiration, helping to combat the Urban Heat Island (UHI) effect—a phenomenon where urban regions experience significantly warmer temperatures than surrounding rural areas. Trees also capture particulate matter and improve air quality, reduce noise pollution, intercept stormwater, and sequester carbon dioxide. Socially, the presence of trees is linked to improved mental well-being, increased property values, and enhanced aesthetic appeal of neighborhoods. More importantly, UTC contributes to climate resilience by buffering heatwaves, reducing energy consumption, and fostering ecological connectivity in fragmented urban landscapes. These contributions position UTC at the core of sustainable urban planning and environmental equity.

Assessing UTC is not only important for quantifying current coverage but also for identifying areas that are underserved and vulnerable to environmental stressors. Such assessments become especially urgent in cities like New York, where the distribution of canopy cover often reflects historical socio-economic and racial inequities. Disadvantaged neighborhoods frequently have fewer trees and are simultaneously more exposed to higher temperatures and poor air quality. A data-driven approach to UTC assessment can therefore inform equitable policy interventions and targeted greening efforts.

1.3 Role of GIS and Remote Sensing in Tree Canopy Mapping

Geographic Information Systems (GIS) and remote sensing technologies play a foundational role in the analysis and mapping of urban tree canopy. These tools enable the collection, visualization, and interpretation of spatial data over large geographic areas with high precision and efficiency. Remote sensing provides multi-temporal satellite imagery and aerial data that capture surface

reflectance, vegetation health, and thermal characteristics. NDVI, derived from Sentinel-2 data, measures vegetation vigor by analyzing the difference between near-infrared and red spectral bands. This index is instrumental in identifying and quantifying vegetation cover across urban landscapes.

Additionally, thermal infrared bands from Landsat 8 are used to derive Land Surface Temperature (LST), helping to pinpoint areas affected by urban heat islands. LiDAR (Light Detection and Ranging), an active remote sensing technology, adds vertical structure to the analysis by capturing 3D information about vegetation height and canopy structure. When processed through software like eCognition or ArcGIS Pro, LiDAR data supports object-based classification, improving the delineation of tree cover from other land cover types such as buildings or bare ground.

GIS platforms integrate these datasets to support spatial modeling, classification, and map production. They allow users to overlay vector layers (e.g., buildings, roads, parks) with raster layers (NDVI, LST, LiDAR-derived elevation) to conduct advanced geospatial analyses. This integration facilitates the identification of spatial patterns, inequities in canopy distribution, and correlations with environmental or socio-economic variables. Overall, GIS and remote sensing offer a comprehensive framework for assessing urban green infrastructure and guiding city-scale environmental planning.

1.4 Objectives of the Study

This study focuses on assessing and mapping UTC coverage in New York City for the year 2021 using GIS, remote sensing, and spatial regression techniques. Specifically, the objectives of this research are fourfold. First, to generate NDVI and LST maps using Landsat 8 satellite data, which will provide spatial insights into the distribution of vegetation and urban heat across NYC. Second, to extract and quantify the proportion of land area covered by tree canopy using a threshold-based classification method. Third, to statistically evaluate the relationship between tree canopy coverage and LST using Ordinary Least Squares (OLS) regression. Fourth, to address spatial dependency in the data through the application of advanced spatial regression models, namely the Spatial Lag Model (SLM) and the Spatial Error Model (SEM), and to test for spatial autocorrelation using Moran's I statistic.

By restricting the analysis to 2021, this study aims to provide a high-resolution, one-year snapshot that captures the immediate state of UTC and its relationship to urban temperature. The findings from this research will help highlight potential hotspots, identify canopy-deficient areas, and contribute to urban forestry strategies that prioritize climate resilience and environmental justice. Ultimately, this project demonstrates the power of geospatial technologies in addressing pressing urban sustainability challenges and offers a template for data-informed policy development in metropolitan contexts.

2. BACKGROUND AND LITERATURE REVIEW

2.1 Definitions and Significance

Urban Tree Canopy (UTC) is defined as the layer of leaves, branches, and stems of trees that cover the ground when viewed from above. It is a critical component of green infrastructure, which includes a network of natural and semi-natural features designed to deliver environmental services such as stormwater management, air and water purification, and biodiversity conservation. The concept of green infrastructure positions UTC as more than an aesthetic feature—it is now recognized as an essential urban asset that contributes to environmental resilience, social well-being, and economic prosperity. In dense cities like New York, where impervious surfaces dominate, the value of UTC is heightened by its ability to mitigate environmental stressors and enhance urban livability.

2.2 Benefits of Urban Tree Canopy

The benefits of UTC span environmental, social, and economic dimensions. Environmentally, trees intercept rainfall, reducing surface runoff and flood risks. They sequester carbon, lower air temperatures through evapotranspiration, and remove air pollutants such as ozone, nitrogen dioxide, and particulate matter. This is particularly relevant in cities facing the dual threats of climate change and air quality degradation. Tree roots also enhance soil quality and reduce erosion, further contributing to water quality improvement.

Socially, access to green spaces with tree canopy improves public health outcomes, reduces stress, encourages physical activity, and is linked to lower crime rates. Green, tree-lined streets have been shown to foster stronger neighborhood connections, enhance mental health, and provide restorative environments that mitigate the psychological impacts of urban living. Trees also act as natural sound buffers, reducing noise pollution in busy urban environments, which has measurable effects on quality of life.

Economically, higher levels of tree canopy coverage are associated with increased property values, especially in residential neighborhoods. Businesses located on greener streets tend to experience

greater foot traffic and consumer satisfaction. Additionally, UTC helps reduce building energy costs by providing shade in summer and windbreak in winter, which contributes to lower greenhouse gas emissions from heating and cooling systems. Research from the U.S. Forest Service and studies like those by Nowak and Greenfield (2018) confirm the significant economic valuation of urban trees, often amounting to millions of dollars in ecosystem services annually for large cities.

From a planning perspective, incorporating UTC into zoning regulations, development incentives, and sustainability plans enhances long-term urban resilience. Tree canopy acts as a visual and spatial marker of community investment and environmental equity, especially when distributed equitably across neighborhoods. In underserved or historically marginalized areas, increasing UTC can support environmental justice goals, reduce health disparities, and contribute to climate adaptation strategies.

Overall, the multi-functional benefits of UTC establish it as an indispensable element of sustainable urban development. As urban populations continue to grow and densify, maintaining and expanding tree canopy will be critical to ensuring livable, inclusive, and climate-resilient cities.



2.3 Literature and Past Studies

Numerous studies have emphasized the importance of accurate UTC mapping and monitoring. The Davey Resource Group has conducted tree canopy assessments in over 150 communities, demonstrating that detailed spatial data can guide targeted planting efforts and improve urban forestry management. A seminal study by MacFaden et al. (2012) conducted for New York City utilized high-resolution LiDAR data and Object-Based Image Analysis (OBIA) to produce a citywide land cover map with per-pixel accuracy exceeding 95%. Their methodology used contextual analysis and rule-based classification in eCognition, enabling fine-scale distinction between tree canopy, buildings, and other urban features. The study's findings influenced NYC Parks' urban forestry initiatives and served as a model for replication in other metropolitan regions.

Moreover, MacFaden et al. noted the limitations of using moderate-resolution imagery such as Landsat for fine-scale urban analysis, advocating instead for the integration of LiDAR and multispectral data. Their workflow included creating a normalized Digital Surface Model (nDSM), Z Deviation layers for texture, and NDVI for vegetation delineation. Such multi-dimensional classification has been instrumental in improving accuracy and realism in tree canopy assessments. This methodology was foundational to our 2021 study, which builds upon their OBIA-LiDAR approach.

2.4 Role of GIS and Remote Sensing

Remote sensing technologies, particularly satellite-based indices and LiDAR have revolutionized UTC analysis. NDVI (Normalized Difference Vegetation Index) is the most widely used vegetation index, capturing differences in near-infrared and red reflectance to measure plant health and density. NDWI (Normalized Difference Water Index) complements NDVI by measuring water content in vegetation. Landsat thermal bands provide Land Surface Temperature (LST), which can be spatially correlated with canopy cover to assess the urban heat island effect. The use of Sentinel-2 imagery offers high temporal and spectral resolution for vegetation monitoring.

LiDAR, a laser-based active sensing technology, offers three-dimensional detail on vegetation height, structure, and texture. Unlike passive sensors, LiDAR is not affected by shadowing and can penetrate canopy layers to provide accurate elevation models. In New York City, where

building density and verticality complicate remote sensing, LiDAR's ability to separate tree canopy from built structures is especially valuable. When integrated in a GIS environment, these datasets allow for advanced spatial modeling, hot-spot detection, and predictive analytics that inform urban greening strategies.

2.5 Urban Greening Policies and Initiatives

Urban forestry has gained traction as a planning priority across U.S. cities. New York City's "MillionTreesNYC" initiative, launched in 2007 under PlaNYC, aimed to plant one million trees by 2017—a goal achieved two years ahead of schedule. This program demonstrated the feasibility of large-scale urban forestry through a blend of municipal funding, private partnerships, and community involvement. Trees were prioritized in low-canopy neighborhoods, addressing environmental justice concerns.

At the national level, the USDA Forest Service's Urban and Community Forestry Program supports cities with grants, technical assistance, and policy guidance. The U.S. Environmental Protection Agency (EPA) has also emphasized green infrastructure as a strategy for stormwater management under the Clean Water Act. These frameworks underscore the growing consensus around tree canopy as a public good and a climate adaptation necessity. With cities facing increasing climate vulnerability, investments in urban greening are now being tied to broader sustainability and equity goals.

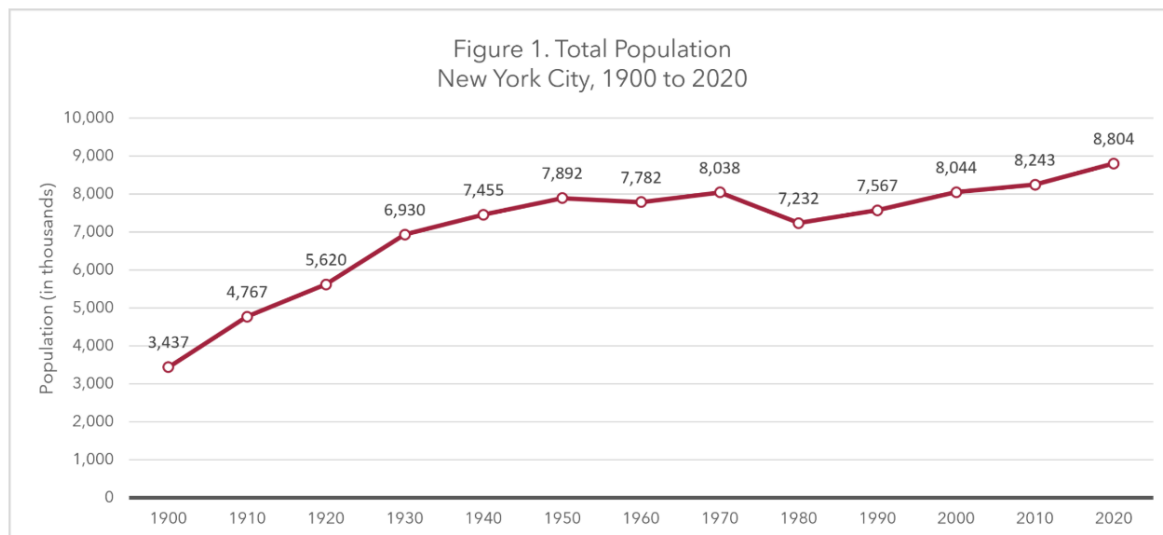
In summary, the literature confirms that urban tree canopy is both an ecological necessity and a spatial justice imperative. Its accurate mapping, especially using GIS and remote sensing, is fundamental to resilient city planning. This project, through its 2021 focus on NYC, adds to this evolving body of knowledge by providing high-resolution analysis that can support data-driven greening interventions.

3. Study Area: Why New York City?

3.1 Population Growth and Urbanization Patterns

New York City (NYC) is one of the most populous and densely built metropolitan areas in the United States. As of the 2020 Census, NYC's population reached approximately 8.8 million people, continuing a long-term upward trend from earlier decades. Historical census data indicates a steady rise from 3.4 million in 1900 to over 7.9 million by 1950, followed by a temporary decline and subsequent growth. This growth has resulted in increasing pressure on urban land use, with much of the city's landscape dominated by built infrastructure, high-rise buildings, and transportation corridors.

Rapid population growth has fueled expansive urbanization, often at the expense of green space. NYC's five boroughs—Manhattan, Brooklyn, Queens, The Bronx, and Staten Island—vary significantly in land use, density, and canopy coverage. Manhattan and Brooklyn, for example, feature intense land development and high impervious surface ratios, limiting the available area for natural vegetation. In contrast, Staten Island and parts of Queens maintain larger green spaces and forested patches. The spatial variability in land cover presents a unique challenge for urban forestry planning and equity in green infrastructure access.



3.2 Urban Heat Island Effects and Green Space Distribution

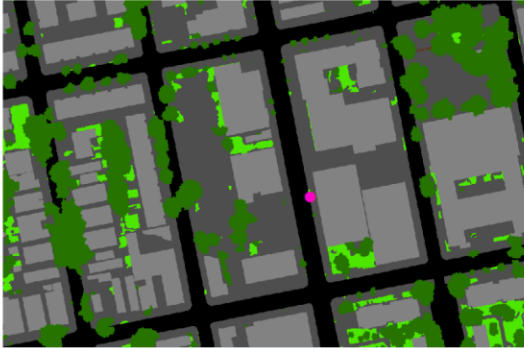
The Urban Heat Island (UHI) effect is one of the most prominent environmental concerns in NYC, exacerbated by dense urbanization and low vegetative cover in many neighborhoods. Surface temperature analysis using Landsat 8 thermal imagery from 2021 indicates a stark contrast between heat-intensive built-up zones and cooler areas with higher tree canopy and vegetated surfaces. As shown in the surface temperature maps and NDVI overlays, boroughs like The Bronx and Brooklyn exhibit significantly different heat profiles based on tree canopy presence.

For instance, Edenwald in The Bronx, which has greater canopy coverage, records lower surface temperatures than Brownsville in Brooklyn, a neighborhood with sparse tree cover and higher levels of impervious infrastructure. These microclimatic differences underline the importance of urban greenery in mitigating localized climate extremes. Green spaces not only cool surrounding areas but also enhance air quality, support biodiversity, and serve as critical recreational and mental health resources.

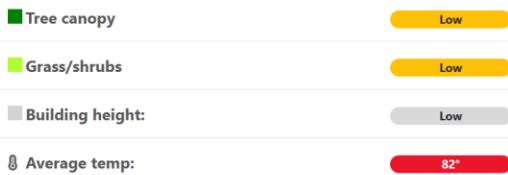
Despite numerous urban greening initiatives, disparities remain. Lower-income communities, often comprised of historically marginalized populations, tend to lack sufficient canopy cover. This spatial inequity limits residents' access to the ecosystem services that tree canopy provides. These neighborhoods bear a disproportionate burden of heat exposure, asthma incidence, and environmental stress. Addressing these disparities requires precise, localized canopy assessments that can guide targeted planting and green infrastructure expansion.

New York City has responded to this challenge with various policies, including the "MillionTreesNYC" initiative, which successfully planted over one million trees across the five boroughs. However, without ongoing assessment and strategic expansion of green infrastructure, these efforts may fall short in sustaining equitable environmental outcomes. The 2021 LiDAR-based canopy analysis serves as a critical benchmark for evaluating the effectiveness of current policies and identifying zones of vulnerability and opportunity.

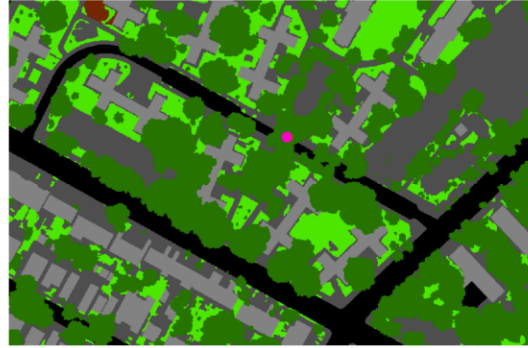
Brownsville, Brooklyn



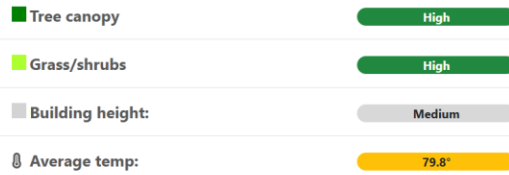
Measure:



Edenwald, the Bronx



Measure:



Tree canopy distribution and UHI variation across New York City. Areas with higher canopy (e.g., Edenwald) show significantly lower surface temperatures compared to less vegetated neighborhoods like Brownsville.

3.3 Problem Statement

As urbanization intensifies, natural landscapes are increasingly converted into built-up environments, leading to a significant loss of green space in metropolitan areas. New York City, as one of the most densely populated urban centers in the United States, exemplifies this trend. High-rise buildings, impervious surfaces, and expanding infrastructure have reduced the availability of land for vegetation and tree canopy coverage. This loss not only affects environmental quality but also undermines climate adaptation and urban resilience strategies.

One of the most pressing consequences of reduced tree cover is the exacerbation of the Urban Heat Island (UHI) effect. Urban areas experience significantly higher temperatures than surrounding rural areas due to heat absorption and radiation from buildings, roads, and other artificial surfaces. In NYC, satellite-derived Land Surface Temperature (LST) maps for 2021 reveal that neighborhoods with low vegetation cover correspond to heat-prone zones, often exceeding surface temperatures by several degrees Celsius compared to greener areas. This thermal imbalance stresses urban infrastructure, increases energy consumption for cooling, and poses serious public

health risks, particularly for vulnerable populations such as the elderly and low-income households.

Furthermore, disparities in tree canopy distribution reveal social inequities in environmental access. Studies show that disadvantaged neighborhoods—often characterized by lower income, higher minority populations, and limited access to green space—consistently register lower UTC percentages. These communities, already burdened by environmental hazards and public health vulnerabilities, face additional challenges from heat exposure and poor air quality. Addressing such environmental justice gaps requires updated, high-resolution assessments of existing tree canopy, especially in underserved boroughs such as the Bronx and central Brooklyn.

Accurate and up-to-date tree canopy mapping is thus critical to supporting data-driven urban forestry initiatives. Traditional vegetation inventories and field surveys, while useful, often lack the resolution or coverage needed for neighborhood-scale planning. Remote sensing and GIS technologies provide the tools necessary to map, monitor, and analyze UTC across the entire urban landscape. In NYC, a comprehensive 2021 analysis using NDVI, LiDAR, and OBIA methods enables the identification of both canopy-rich and canopy-deficient areas, guiding targeted planting strategies and long-term green infrastructure investments.

4. Data Sources

The analysis of urban tree canopy and its relationship with land surface temperature in New York City was based entirely on data from the year 2021. The following geospatial and statistical datasets were utilized:

4.1 Satellite Imagery and Indices (2021)

- **Sentinel-2 (2021):** Used to derive the Normalized Difference Vegetation Index (NDVI) and Normalized Difference Built-up Index (NDBI) for vegetation health and urban structure assessment.
- **Landsat 8 (2021):** Thermal bands were used to calculate Land Surface Temperature (LST) across New York City.
- **Derived Indices:** NDVI, NDWI (Normalized Difference Water Index), and NDBI were processed using Google Earth Engine and visualized using QGIS and ArcGIS Pro.

4.2 Raster and Vector Data

- **NYC Boundary Shapefiles:** Provided base layers for administrative borders and masking.
- **Land Cover and Tree Canopy Masks:** Derived from classification maps using NDVI thresholds and manual correction in QGIS.
- **Bus Stops, Parks, and Road Layers:** Included to support overlay analysis of accessibility and socio-environmental context.

4.3 Spatial Analytical Tools

- **Google Earth Engine (GEE):** Used for pre-processing satellite data and generating indices.
- **Python Libraries:** GeoPandas, Rasterio, PySAL, and Matplotlib were used for spatial correlation analysis (e.g., Moran's I, LISA) and regression modeling.
- **ArcGIS Pro:** For visual map composition, layer integration, and classification-based overlays.

4.4 Statistical & Modeling Data

- **OLS Regression Data (2021):** Socio-demographic and environmental variables were compiled at the neighborhood level.
- **SLM and SEM Models:** Conducted using spatial weights matrices to account for spatial dependence.
- **Geographically Weighted Regression (GWR):** Used to assess spatial heterogeneity in canopy–LST relationships.

This integrated dataset architecture facilitated a multi-scale analysis of tree canopy distribution and its ecological significance, with a specific focus on thermal stress, equity, and resilience in New York City.

5. Methodology

This study employs a structured spatial analysis approach using remote sensing, GIS, and statistical modeling tools to evaluate urban tree canopy (UTC) and its relationship with Land Surface Temperature (LST) in New York City for the year 2021.

5.1 Data Collection

- **Primary Data Sources:**
 - **Sentinel-2 imagery (2021)** for NDVI and NDBI computation
 - **Landsat 8 thermal imagery (2021)** for LST calculation
- **Vector Layers:** NYC borough boundaries, parks, roads, and tree canopy masks sourced from NYC Open Data
- **Derived Layers:** NDVI and LST rasters were generated and processed using Google Earth Engine (GEE)

5.2 Data Processing and Analysis

- **Software Platforms:**
 - **Google Earth Engine (GEE)** for spectral index calculation (NDVI, NDBI) and LST extraction
 - **QGIS and ArcGIS Pro** for spatial overlay, classification, and thematic mapping
 - **Python libraries** (GeoPandas, Rasterio, PySAL) for spatial statistics and regression modeling
- **Analytical Techniques:**
 - **NDVI Mapping:** Used to identify and quantify vegetation density
 - **NDBI Mapping:** Helped distinguish built-up areas with low vegetation
 - **Land Surface Temperature (LST):** Derived using Landsat 8 Band 10 and converted via mono-window algorithm

- **Spatial Autocorrelation:** Moran's I and LISA cluster analysis to detect spatial clustering in LST and canopy cover
- **Regression Analysis:**
 - **OLS (Ordinary Least Squares)** to establish baseline relationships
 - **Spatial Lag Model (SLM)** and **Spatial Error Model (SEM)** to account for spatial dependence
 - **Geographically Weighted Regression (GWR)** to explore localized variation in canopy–temperature relationships

5.3 Visualization and Output Maps

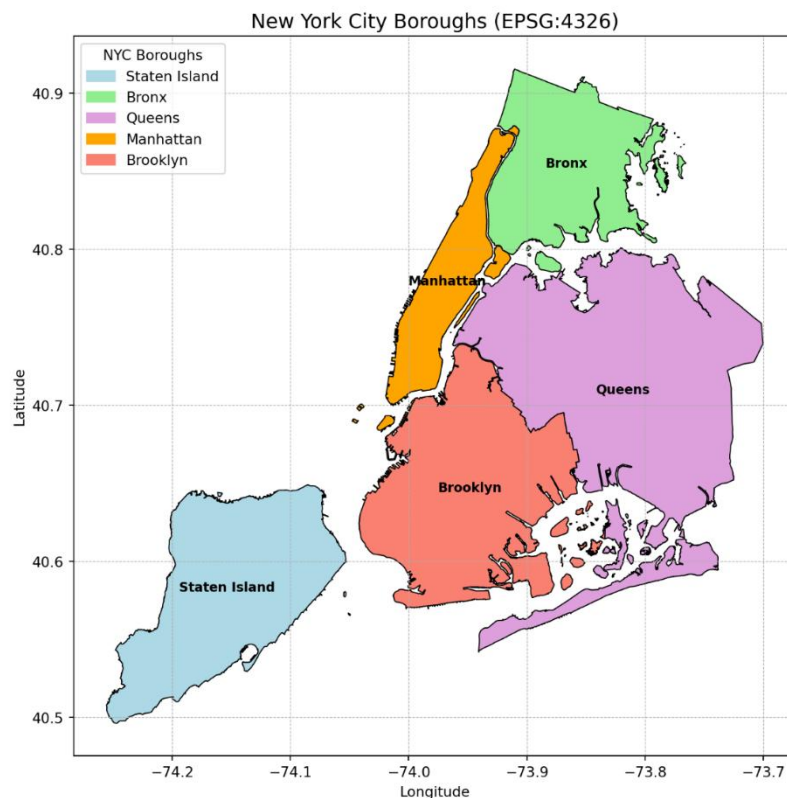
- **Key Maps Produced (2021):**
 - NDVI Classification Map
 - Land Surface Temperature Map
 - Tree Canopy Hotspot (G_i^*) Analysis
 - LISA Cluster Maps for LST and Canopy
 - OLS, SLM, and SEM Residual Maps
 - GWR Coefficient Map

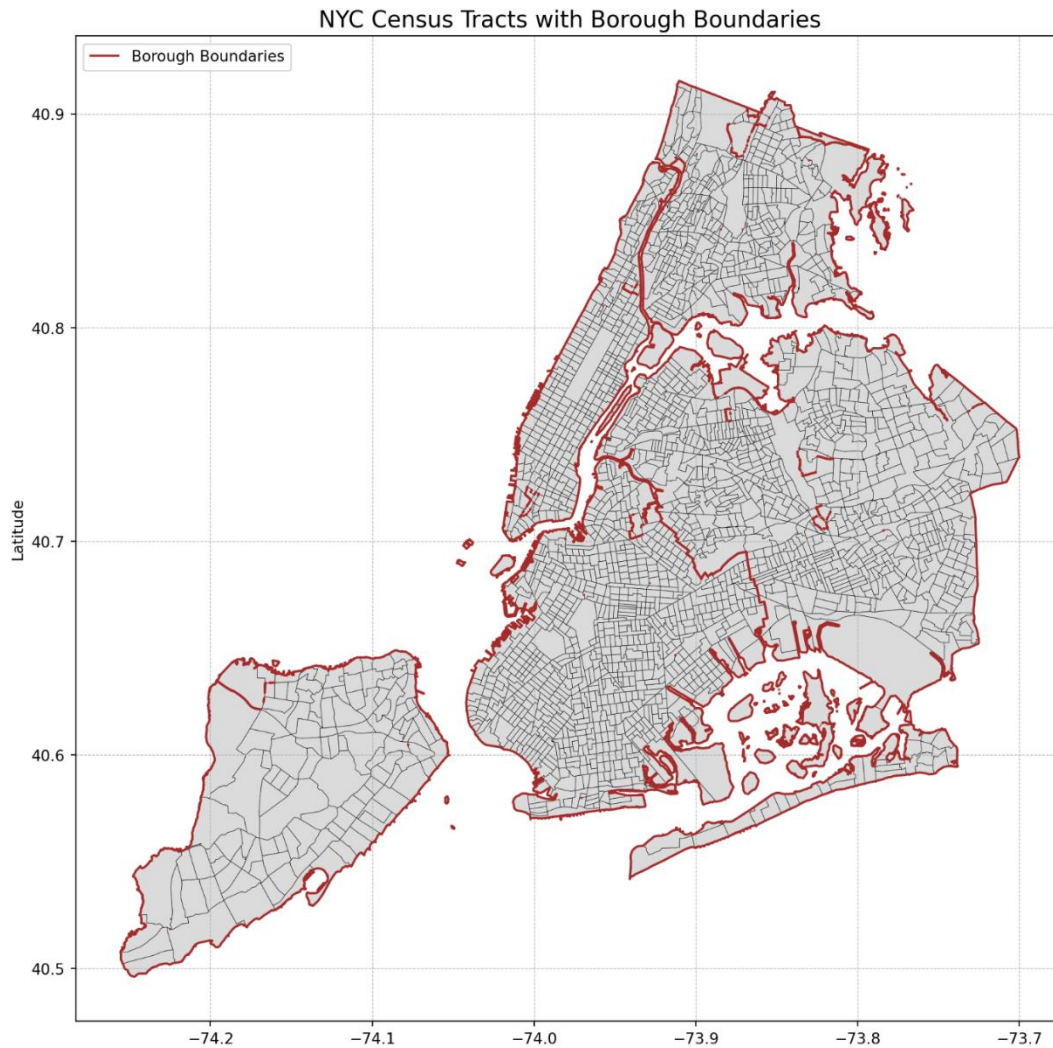
6. Data Collection

6.1 Spatial Context:

New York City Boroughs and Census Tracts

To begin the analysis, borough-level and tract-level administrative boundaries were mapped using shapefiles from NYC Open Data. These maps provide the geographical framework for all subsequent layers. The borough map serves as a foundational spatial reference that illustrates the extent and layout of New York City's five boroughs—Bronx, Brooklyn, Manhattan, Queens, and Staten Island. Each borough exhibits a unique urban morphology and landscape composition, which directly affects tree canopy presence and vegetation patterns. For example, Staten Island, known for its suburban and low-density development, demonstrates significantly higher levels of vegetation and tree coverage. In contrast, Manhattan's dense built environment, dominated by vertical infrastructure and commercial activity, shows a lower propensity for green cover. These geographical distinctions are vital for understanding tree canopy disparities, especially when comparing urban density with vegetation indices.

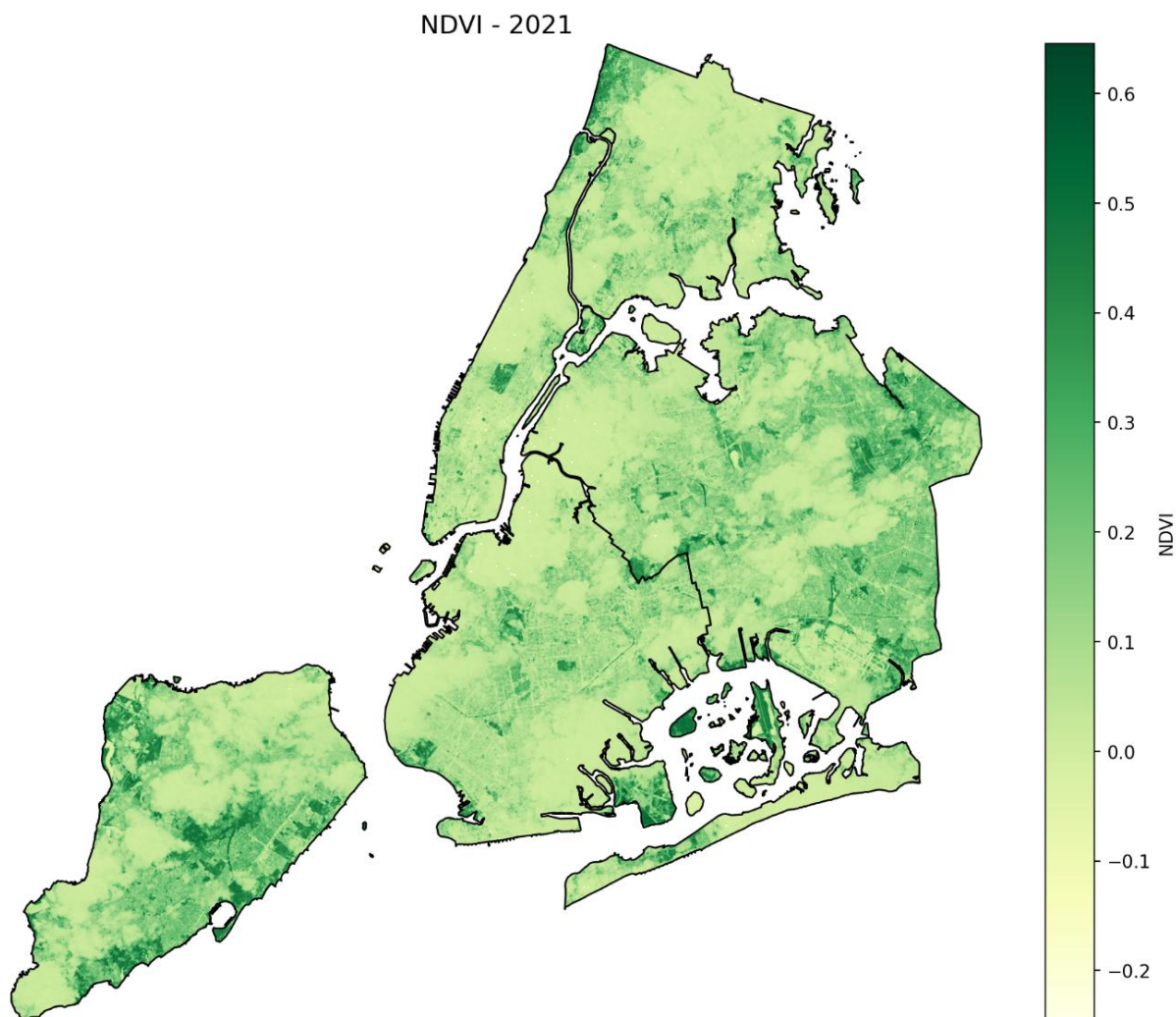




6.2 Vegetation and Temperature Analysis (2021)

6.2.1 NDVI – Normalized Difference Vegetation Index (2021)

NDVI was computed from Sentinel-2 imagery and is a well-established vegetation index that distinguishes healthy vegetation from non-vegetated surfaces using reflectance in the red and near-infrared bands. The NDVI 2021 map for NYC reveals clear vegetative gradients across boroughs. Staten Island, Bronx, and select portions of Queens exhibit high NDVI values (0.4 to 0.6), indicative of tree-dense residential areas and parklands. Central Brooklyn and western Queens show mid-level NDVI values, while parts of Manhattan and industrial sections of the waterfront record the lowest NDVI values (<0.1), indicating impervious surface dominance.



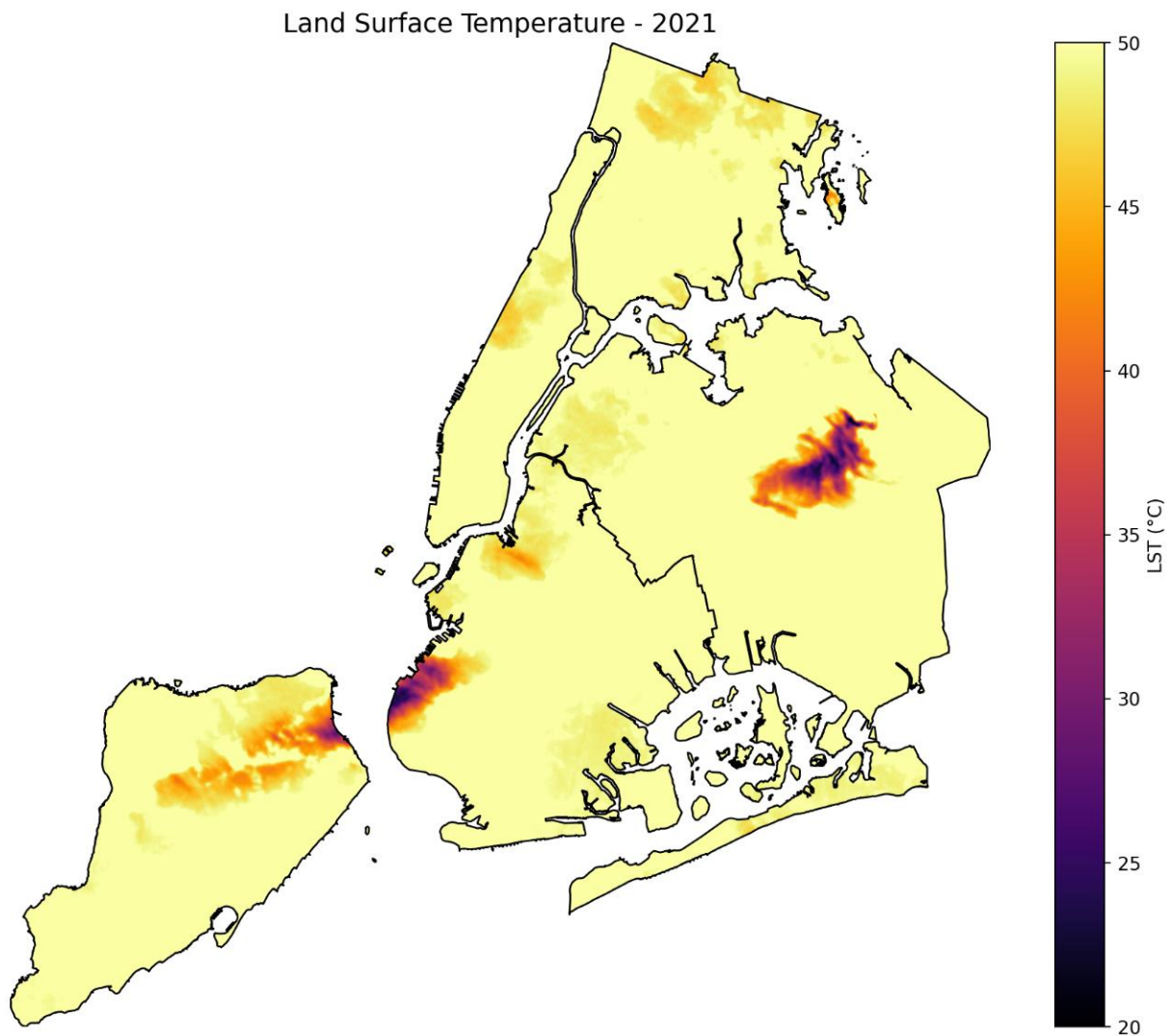
NDVI map highlighting vegetative density. Values range from -0.2 to +0.6.

NDVI not only maps greenness but also aids in quantifying tree canopy extents. Higher NDVI zones strongly correlate with cooler surface temperatures, reinforcing vegetation’s role in urban microclimate regulation. This map served as the basis for tree canopy classification and was later used in overlay analyses with LST and demographic datasets to assess green equity.

6.2.2 Land Surface Temperature (LST) – 2021

The LST map highlights the distribution of thermal intensity across NYC, derived from Landsat 8 Band 10 using the mono-window algorithm. Temperature gradients range from 20°C to over 50°C. Hotspots are clearly visible in lower Manhattan, eastern Brooklyn, and central Queens—areas generally characterized by low vegetation and high impervious surface concentrations.

Conversely, cool zones cluster around large parks such as Prospect Park, Central Park, and Pelham Bay Park.



Land Surface Temperature derived from Landsat 8 Band 10 using mono-window algorithm.

Spatial analysis of this map revealed that boroughs with higher NDVI values experienced lower LST. This finding validates the role of vegetation in urban cooling and provides a compelling case for expanding green infrastructure to mitigate heat islands. Moreover, comparing LST patterns with demographic layers showed that vulnerable communities, often with lower canopy coverage, are disproportionately affected by urban heat.

6.2.3 LST Summary Statistics

Statistical summaries extracted from the LST raster reinforce spatial trends observed in the map. The temperature ranged from a minimum of 17.5°C to a maximum of 77.1°C, with a mean temperature of approximately 57.5°C. The standard deviation was around 7.91°C, indicating moderate variability in temperature values across census tracts. This variance corresponds closely with differences in urban form and green coverage.

	Value
Min	17.516785
Max	77.155846
Mean	57.537819
Median	58.272125
Std Dev	7.914862

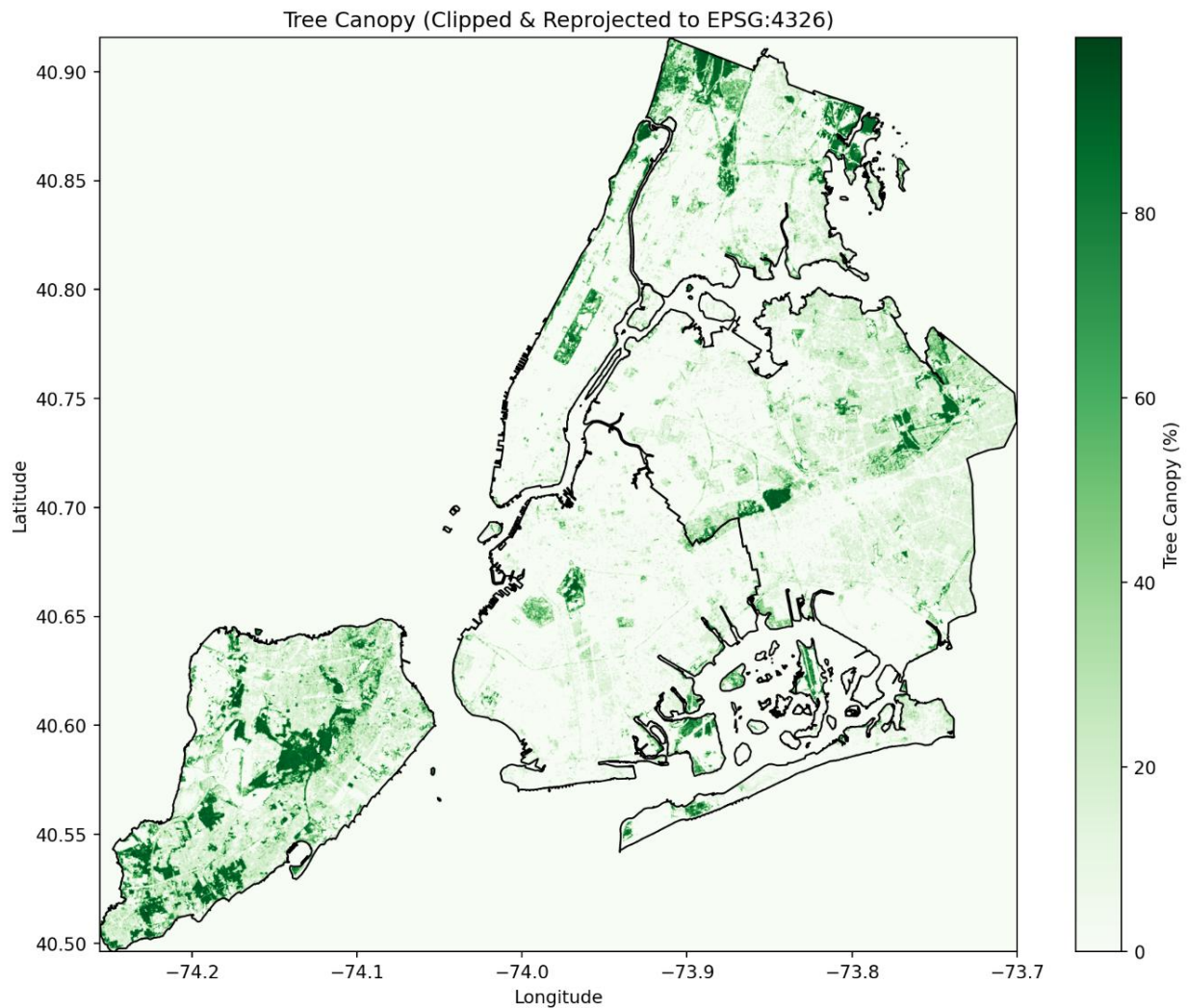
Summary metrics for LST – max: 77.15°C, mean: 57.54°C.

These values highlight the severity of the urban heat problem in NYC, especially during peak summer months. Temperature extremes exacerbate energy demand for cooling and contribute to heat-related health risks, particularly in densely built, tree-deficient neighborhoods. Thus, monitoring LST changes annually is essential for long-term climate adaptation.

6.2.4 Tree Canopy Cover (2011)

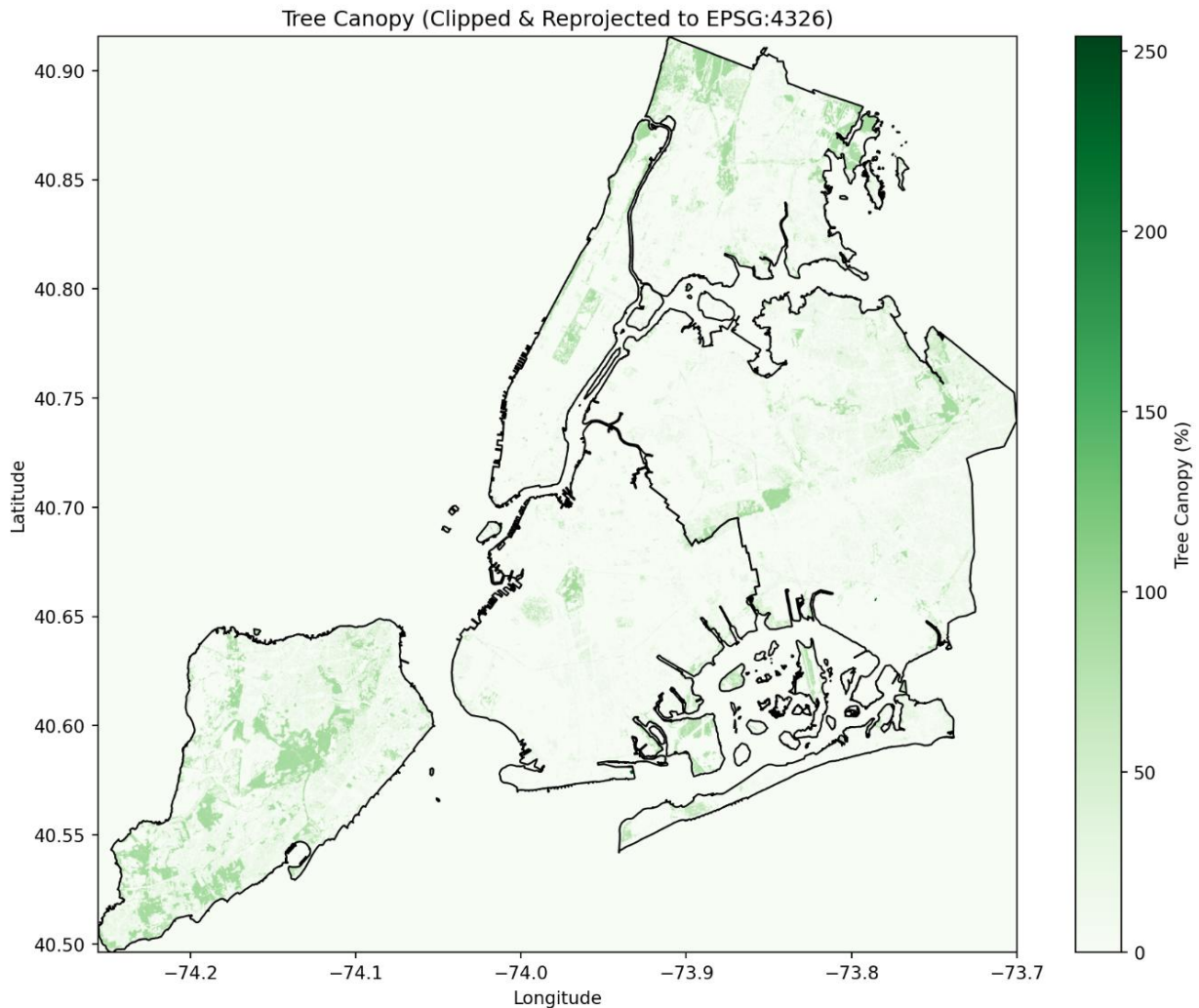
The 2011 tree canopy map provides a baseline assessment of NYC’s green infrastructure before the implementation of major greening initiatives such as MillionTreesNYC. The raster was processed from high-resolution vegetation classification layers and reprojected for consistency. Staten Island and the Bronx already exhibited significant tree cover in 2011, especially in parkland and lower-density residential zones. Manhattan and parts of Brooklyn displayed limited vegetation, constrained by the dominance of built structures.

This layer served as a reference for assessing urban canopy expansion or loss over time. It is particularly valuable for identifying areas that have remained stagnant or regressed despite citywide tree planting goals. The 2011 data highlighted both legacy green zones and historically underserved neighborhoods.



6.2.5 Tree Canopy Cover (2021)

The tree canopy map displays the spatial distribution of vegetated areas identified using NDVI thresholds and classified canopy masks. This output was clipped to NYC boundaries and reprojected to EPSG:4326. Staten Island stands out with the most extensive green cover, attributed to its suburban layout and preserved forest zones. The Bronx and Queens also contain notable green corridors and park systems, while dense sections of Manhattan and central Brooklyn show significant canopy gaps.

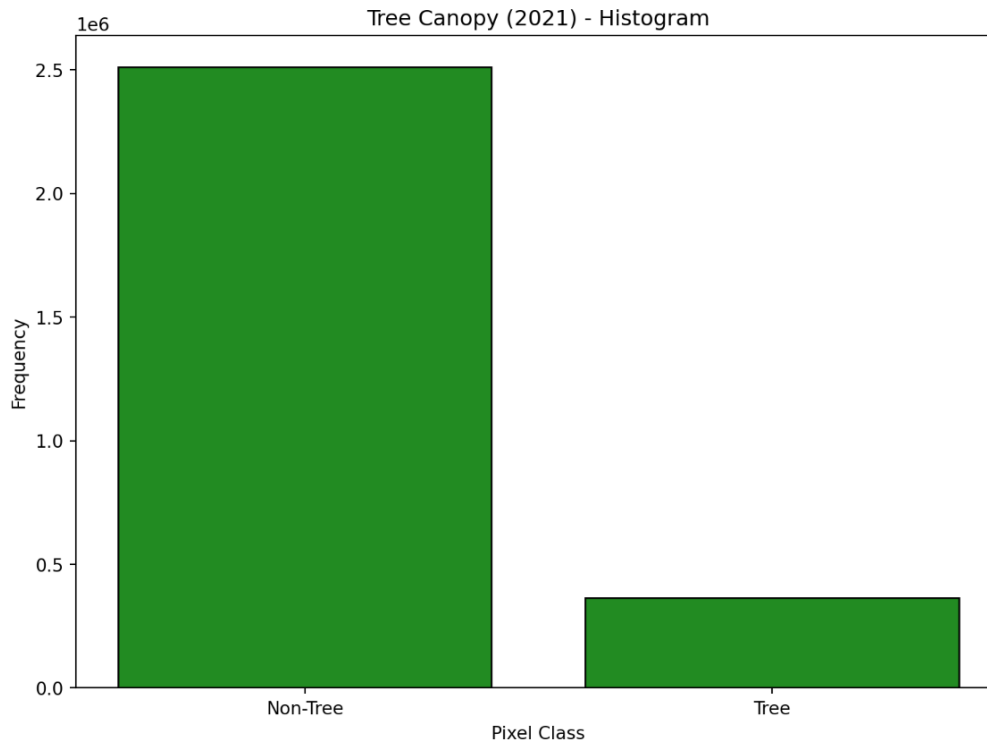


Tree canopy layer reprojected to EPSG:4326. Highest density observed in Staten Island and park-rich Bronx.

This spatial output was vital in detecting which neighborhoods are underserved in canopy provisioning. Additionally, the tree canopy raster was used in spatial correlation and regression models to assess how green infrastructure contributes to heat mitigation and social well-being.

6.2.6 Tree Canopy (2021) Histogram

The canopy histogram depicts pixel distribution between tree and non-tree categories across the city. As shown, non-tree pixels dominate the landscape, emphasizing the high degree of impervious surfaces in NYC. Only a small proportion of total pixels are classified as tree, further highlighting the pressing need for urban reforestation efforts.



🇺🇸 Tree Canopy Classification Summary:

	Value
Total Pixels	2.875296e+06
Tree Pixels	3.632610e+05
Non-Tree Pixels	2.512035e+06
Tree Proportion	1.263386e-01
Non-Tree Proportion	8.736614e-01

Majority of pixels in NYC classified as non-tree.

This histogram helps validate the spatial canopy map and confirms the imbalance in land cover. It also serves as a statistical indicator for setting citywide greening benchmarks and planning interventions.

6.2.7 Decadal Comparison (2011 vs 2021)

A comparison between the 2011 and 2021 tree canopy maps reveals nuanced trends in urban green coverage. While some boroughs have experienced gains due to city-led initiatives, others have shown stagnation or even regression. Staten Island and the northern Bronx saw modest increases

in canopy density, largely around established green corridors and community planting zones. Meanwhile, commercial centers and industrial hubs in Brooklyn and Queens suffered minor losses, likely due to development pressures and insufficient tree preservation policies.

Despite NYC's ambitious MillionTrees campaign and greening programs, the data indicates that overall growth has been uneven. Low-income and densely populated neighborhoods have not benefited proportionately. Several census tracts that experienced extreme heat in 2011 continue to lack adequate green infrastructure. These disparities underscore the need for targeted, equity-based urban forestry planning to ensure sustainable environmental access across all communities.

7. Data Analysis

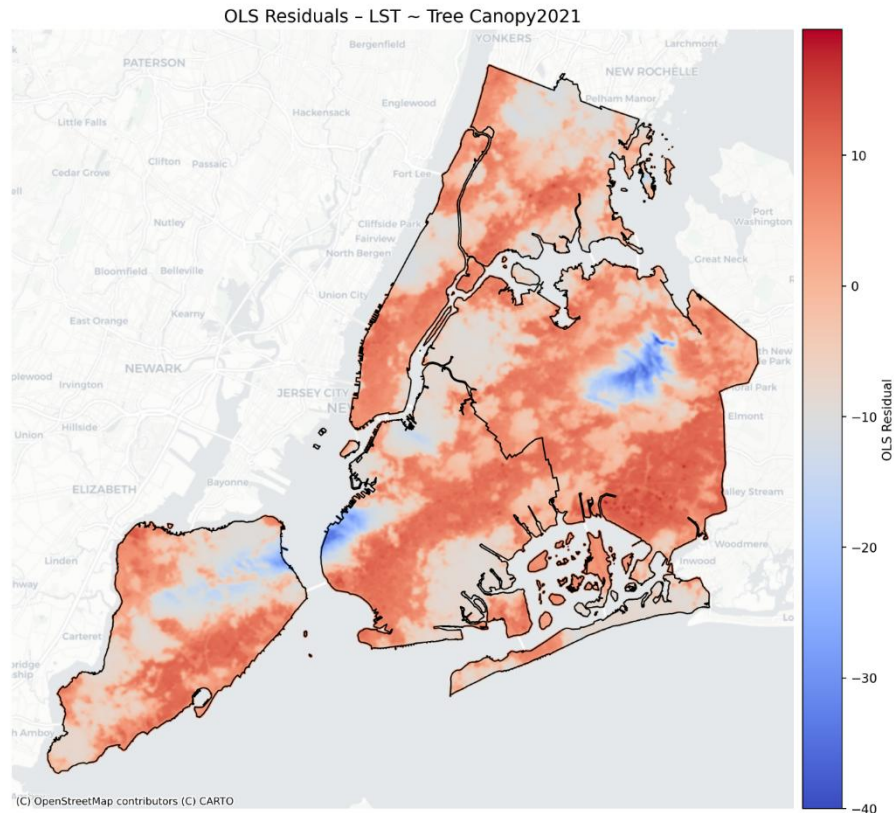
In this phase of the project, a series of statistical and spatial regression techniques were applied to analyze the relationship between urban tree canopy coverage and Land Surface Temperature (LST) across New York City for the year 2021. These methods help to quantify how vegetation affects urban heat distribution and identify spatial disparities that are important for targeted green infrastructure planning. The models used include Ordinary Least Squares (OLS) Regression, Spatial Autocorrelation through Moran's I, Spatial Lag Model (SLM), Spatial Error Model (SEM), Geographically Weighted Regression (GWR), and Hotspot Analysis (Z-score and Getis-Ord Gi*). Each model provides different insights into spatial patterns and dependencies. These models aim to statistically assess the relationship between Land Surface Temperature (LST) and tree canopy coverage, allowing us to identify spatial dependencies and local variations that conventional methods cannot reveal. Understanding these spatial dynamics is essential for informed urban greening policies, equitable tree planting, and climate adaptation planning.

7.1 OLS Regression (2021)

The OLS model was employed to examine the global linear relationship between tree canopy percentage and LST values across the city. The regression results indicated a very weak but statistically significant negative correlation between canopy cover and LST. The coefficient for canopy was -0.0024, suggesting that for each 1% increase in canopy, there is an approximate 0.0024°C decrease in LST. However, the R-squared value was nearly zero, indicating limited explanatory power due to spatial heterogeneity.

```

=====
                        OLS Regression Results
=====
Dep. Variable:          value    R-squared:                0.000
Model:                  OLS      Adj. R-squared:            0.000
Method:                 Least Squares    F-statistic:         14.42
Date:                   Wed, 30 Apr 2025    Prob (F-statistic):    0.000146
Time:                   23:05:22    Log-Likelihood:       -3.0305e+06
No. Observations:      868912    AIC:                  6.061e+06
Df Residuals:          868910    BIC:                  6.061e+06
Df Model:               1
Covariance Type:       nonrobust
=====
                        coef      std err          t      P>|t|      [0.025      0.975]
-----
const          57.5302         0.009    6592.291     0.000     57.513     57.547
canopy          0.0024         0.001       3.797     0.000      0.001     0.004
=====
Omnibus:          42104.722    Durbin-Watson:        0.009
Prob(Omnibus):    0.000    Jarque-Bera (JB):     48464.444
Skew:             -0.570    Prob(JB):              0.00
Kurtosis:         3.200    Cond. No.              14.1
=====
```



Key Results:

- Coefficient for canopy: -0.0024
- R-squared: 0.000
- P-value (canopy): < 0.001 (significant)

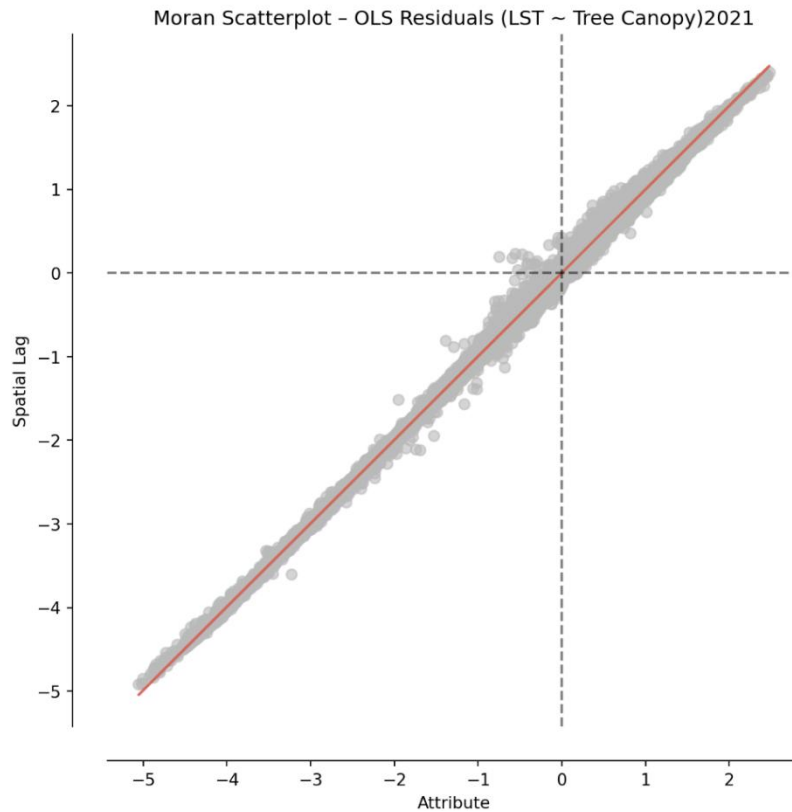
7.2 Moran's I on OLS Residuals (2021)

To test whether the residuals from the OLS model are spatially autocorrelated, a Moran's I test was conducted. The value of Moran's I was 0.9976 with a p-value of 0.001, confirming that spatial dependence remains in the residuals, and that OLS alone is insufficient to model the spatial structure in the data. This justified the use of spatial regression models.

Key Results:

- Moran's I: 0.9976
- P-value: 0.0010

Moran's I: 0.9976
P-value: 0.0010



7.3 Spatial Lag Model (SLM) (2021)

The SLM incorporated spatial dependency into the regression model. The lag term (W_LST) was highly significant, and the model exhibited improved log-likelihood and reduced Akaike Information Criterion (AIC) compared to OLS. However, the canopy variable was not statistically significant in this spatial context.

Key Results:

- Pseudo R-squared: 0.0010
- Coefficient (canopy): 0.00008 (not significant)

REGRESSION RESULTS

SUMMARY OF OUTPUT: MAXIMUM LIKELIHOOD SPATIAL LAG (METHOD = LU)

Data set	: unknown		
Weights matrix	: unknown		
Dependent Variable	: LST	Number of Observations:	5000
Mean dependent var	: 57.5017	Number of Variables	: 3
S.D. dependent var	: 7.9621	Degrees of Freedom	: 4997
Pseudo R-squared	: 0.8878		
Spatial Pseudo R-squared	: 0.0010		
Log likelihood	: -12661.9330		
Sigma-square ML	: 7.4616	Akaike info criterion	: 25329.866
S.E of regression	: 2.7316	Schwarz criterion	: 25349.418

Variable	Coefficient	Std. Error	z-Statistic	Probability
CONSTANT	2.97813	0.21785	13.67063	0.00000
canopy	0.00008	0.00312	0.02639	0.97894
W_LST	0.94802	0.00374	253.58027	0.00000

SPATIAL LAG MODEL IMPACTS

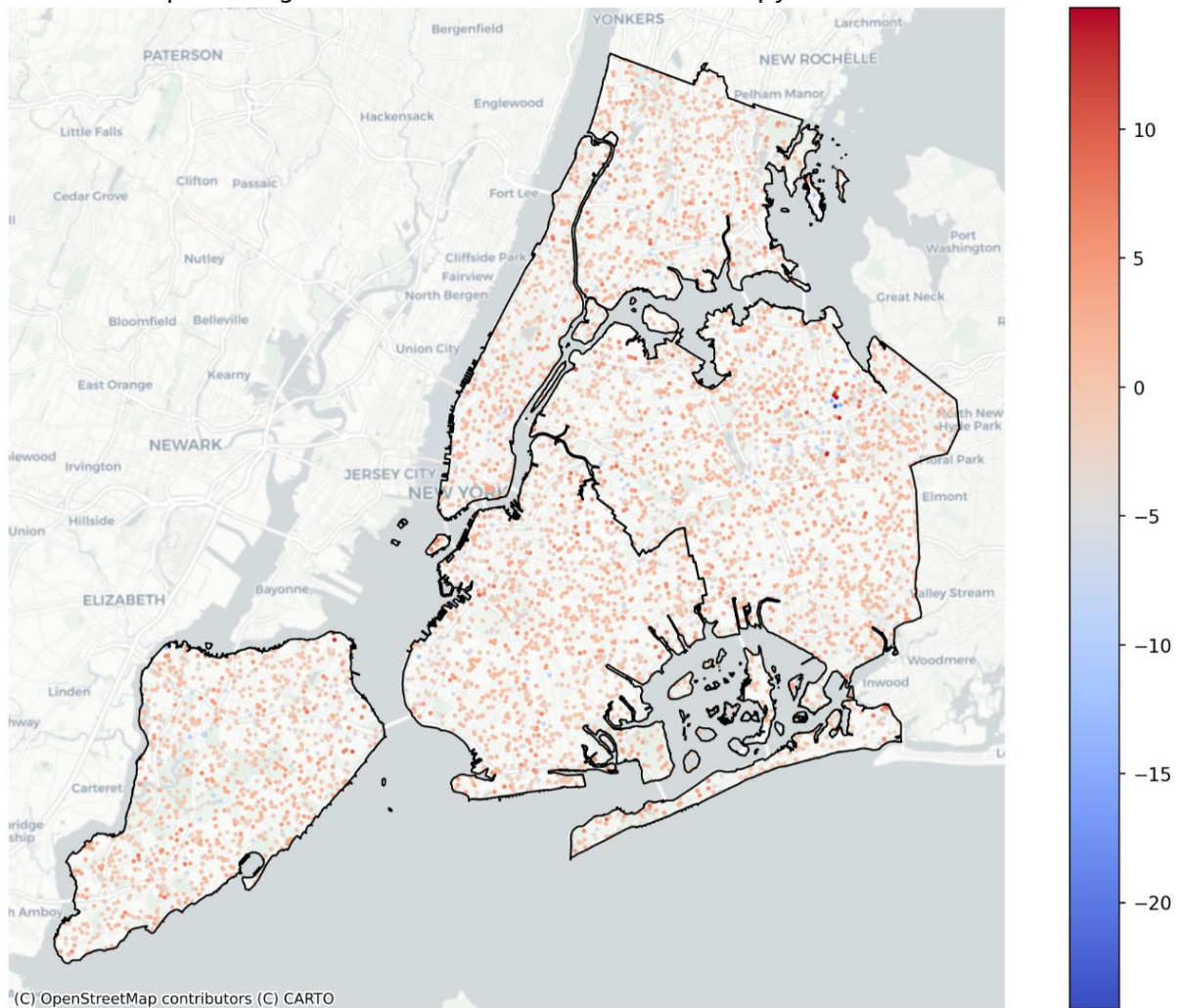
Impacts computed using the 'simple' method.

Variable	Direct	Indirect	Total
canopy	0.0001	0.0015	0.0016

===== END OF REPORT =====

- Significant lag effect (W_LST)

Spatial Lag Model Residuals - LST ~ Tree Canopy 2021



7.4 Spatial Error Model (SEM) (2021)

The SEM addressed spatial autocorrelation through the error term. Similar to the SLM, it improved upon the OLS in model fit. However, the tree canopy variable remained insignificant in this model as well.

Key Results:

- Coefficient (canopy): -0.00181
- Pseudo R-squared: 0.0000
- Significant spatial error component (lambda = 0.94836)

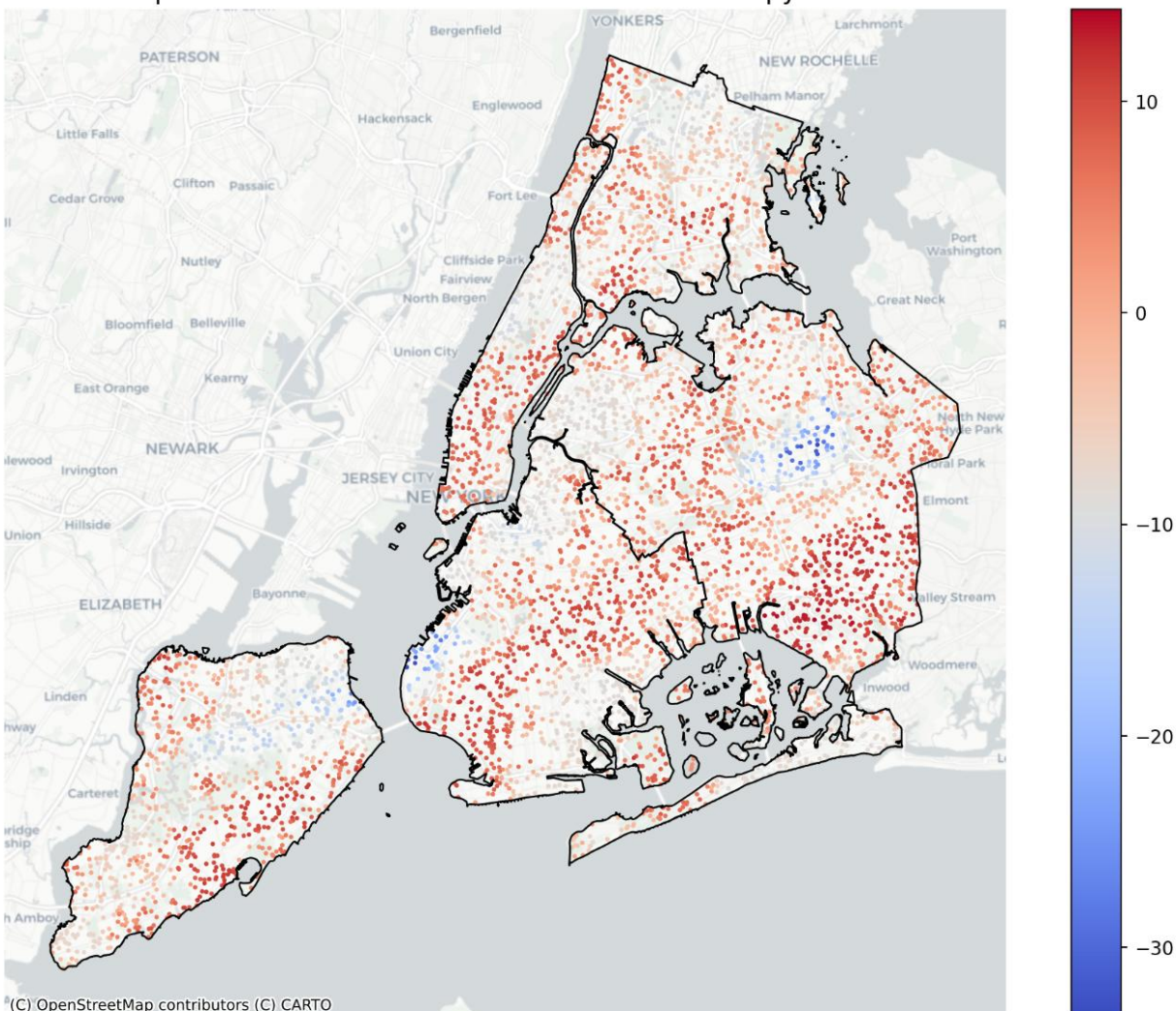
REGRESSION RESULTS

SUMMARY OF OUTPUT: ML SPATIAL ERROR (METHOD = full)

Data set	:	unknown		
Weights matrix	:	unknown		
Dependent Variable	:	LST	Number of Observations:	5000
Mean dependent var	:	57.4966	Number of Variables	2
S.D. dependent var	:	7.8727	Degrees of Freedom	4998
Pseudo R-squared	:	0.0000		
Log likelihood	:	-12639.8939		
Sigma-square ML	:	7.3802	Akaike info criterion	25283.788
S.E of regression	:	2.7167	Schwarz criterion	25296.822

Variable	Coefficient	Std. Error	z-Statistic	Probability
CONSTANT	57.27271	0.74413	76.96634	0.00000
canopy	-0.00181	0.00282	-0.64348	0.51992
lambda	0.94836	0.00365	260.15342	0.00000

Spatial Error Model Residuals - LST ~ Tree Canopy 2021

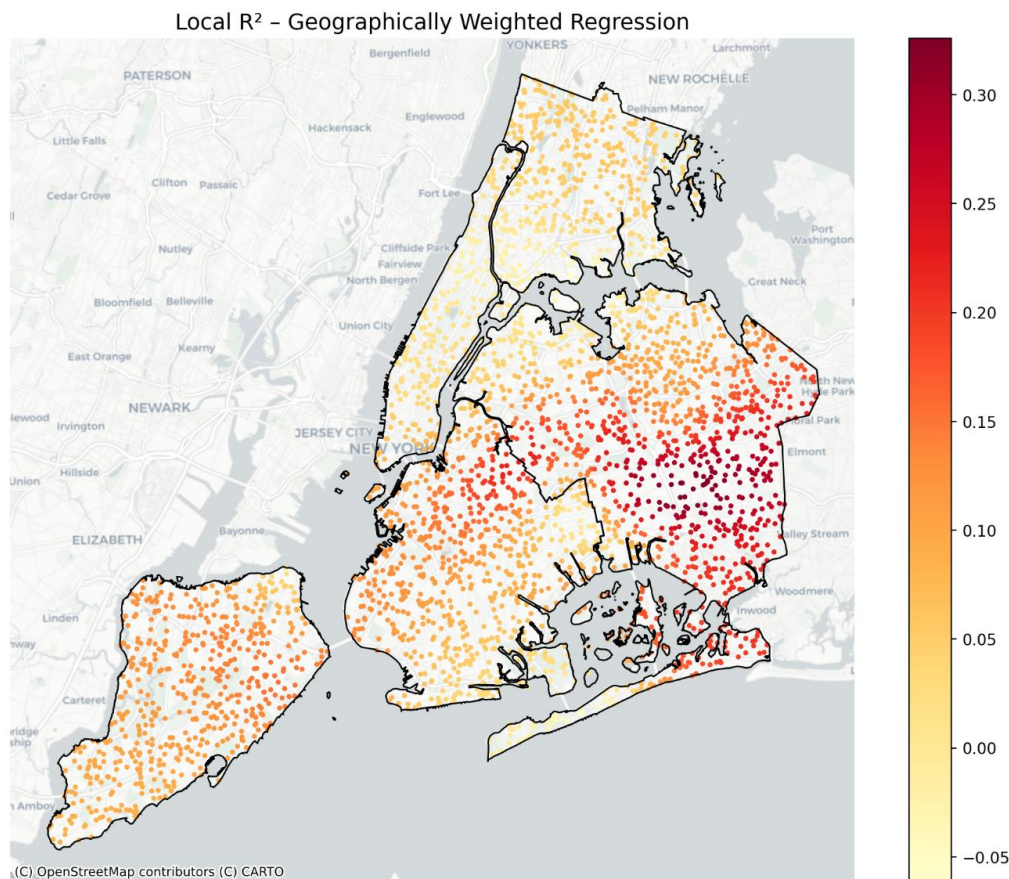


7.5 Geographically Weighted Regression (GWR) (2021)

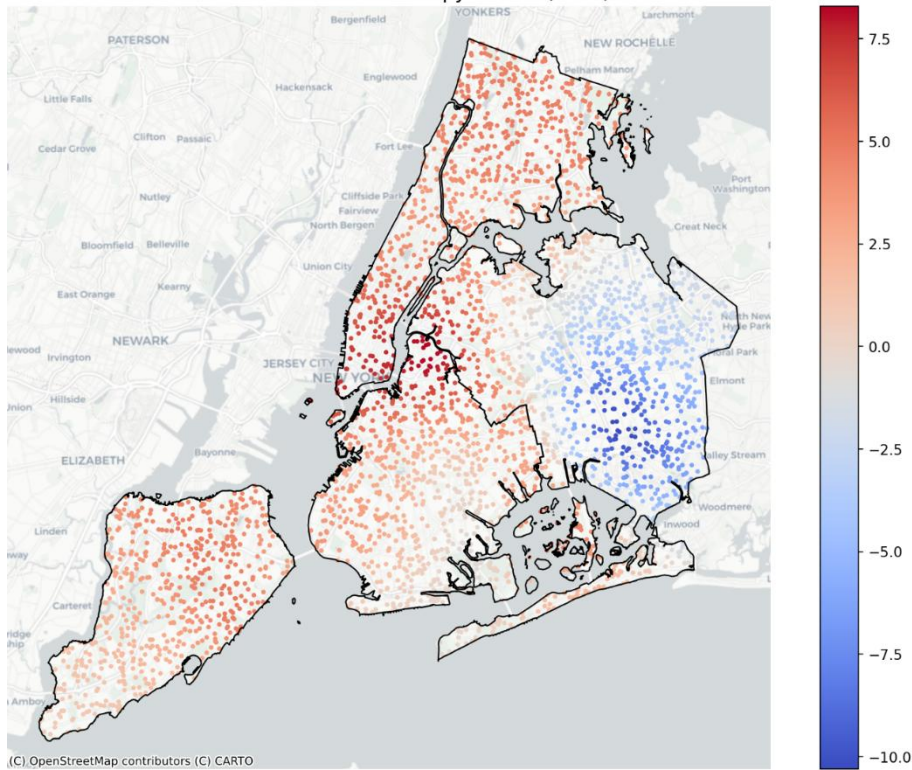
GWR was used to account for spatial non-stationarity in the relationship between tree canopy and LST. The local R-squared values and coefficients varied significantly across the city. Some areas, particularly in Brooklyn and Queens, showed stronger negative effects of canopy on LST. This model offered the most nuanced understanding of spatial variation.

Key Outputs:

- Local R^2 range: ~0.00 to 0.30
- Local coefficients ranged from -10 to +7.5



Local Coefficient - Effect of Canopy on LST (GWR)2021



Global Regression Results

Residual sum of squares:	189128.665
Log-likelihood:	-10472.475
AIC:	20948.950
AICc:	20950.958
BIC:	165117.575
R2:	0.000
Adj. R2:	-0.000

Variable	Est.	SE	t(Est/SE)	p-value
X0	57.557	0.145	396.919	0.000
X1	0.090	0.145	0.622	0.534

Geographically Weighted Regression (GWR) Results

Spatial kernel:	Adaptive bisquare
Bandwidth used:	750.000

Diagnostic information

Residual sum of squares:	159311.628
Effective number of parameters (trace(S)):	19.578
Degree of freedom (n - trace(S)):	2980.422
Sigma estimate:	7.311
Log-likelihood:	-10215.190
AIC:	20471.538
AICc:	20471.836
BIC:	20595.140
R2:	0.158
Adjusted R2:	0.152
Adj. alpha (95%):	0.005
Adj. critical t value (95%):	2.802

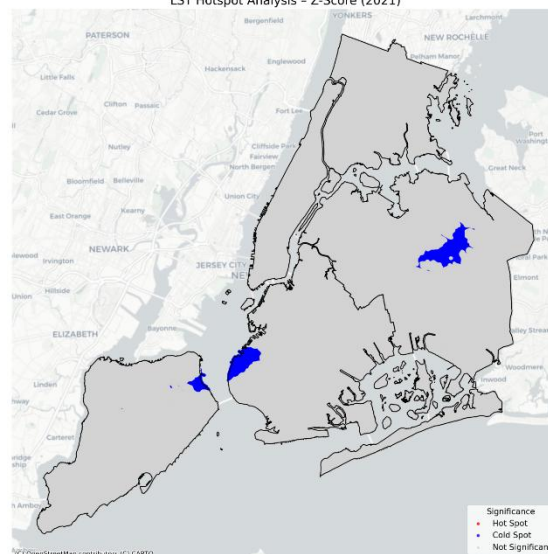
Summary Statistics For GWR Parameter Estimates

Variable	Mean	STD	Min	Median	Max
X0	57.555	1.868	52.483	57.179	62.405
X1	0.127	0.343	-0.898	0.234	0.723

7.6 Hotspot Analysis – LST (Z-score Method) 2021

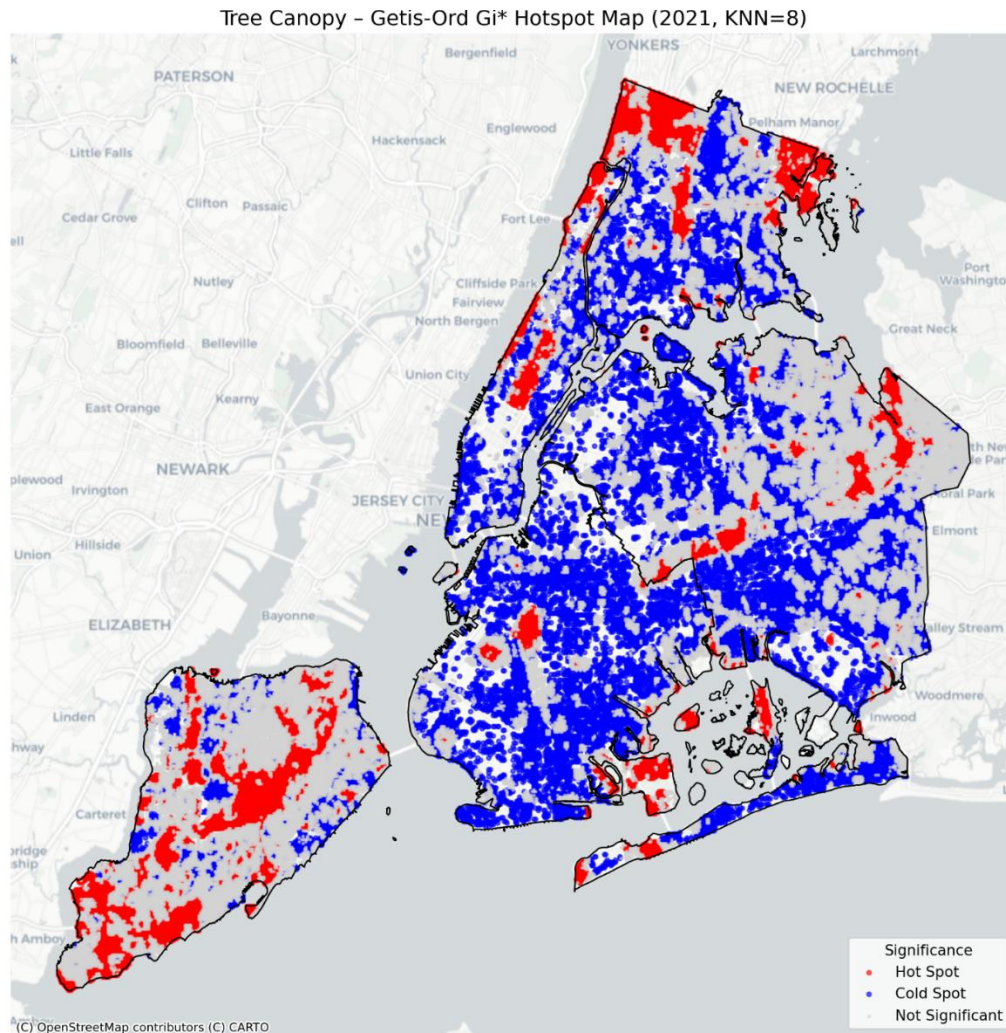
This analysis identified statistically significant hotspots of high LST using the Z-score method. Areas such as southern Brooklyn, eastern Bronx, and western Staten Island emerged as prominent heat hotspots. This visualization helped guide targeted analysis for tree canopy impact.

LST Hotspot Analysis - Z-Score (2021)



7.7 Hotspot Analysis – Tree Canopy (Getis-Ord Gi*) 2021

The Getis-Ord Gi* statistic was applied to identify significant hot and cold spots in tree canopy distribution. Cold spots (areas with significantly lower canopy) were widespread in eastern Queens and central Brooklyn, while hot spots (dense canopy) were mostly in northern Bronx and parts of Staten Island. This further supports equity-focused planning.



8. RESULTS

8.1 Summary of Key Findings

This study analyzed the relationship between urban tree canopy (UTC) and Land Surface Temperature (LST) across New York City in 2021. The following are the primary findings:

- **Tree Canopy Coverage:** NYC's average canopy coverage is approximately 12.63%, with the highest densities observed in Staten Island and the Bronx. In contrast, central Brooklyn and lower Manhattan exhibited the least coverage.
- **Vegetation and Heat Correlation:** NDVI and LST maps indicated a negative correlation between vegetation density and surface temperature. Areas with higher NDVI typically experienced lower LST, confirming the cooling effect of vegetation.
- **OLS Regression:** Demonstrated a weak but significant inverse relationship between canopy percentage and LST (coefficient: -0.0024; $p < 0.001$), though R^2 was very low (~ 0.000), suggesting other influencing factors.
- **Spatial Autocorrelation:** Moran's I value (0.9976) confirmed strong spatial clustering in the residuals of the OLS model, necessitating spatial regression models.
- **Spatial Lag Model & Spatial Error Model:** These showed better statistical fit but still found canopy coverage statistically insignificant in explaining LST, emphasizing spatial complexity in urban heat patterns.
- **GWR Model:** Revealed significant spatial heterogeneity—in areas like Brooklyn and Queens, canopy had a stronger cooling effect than in others, with local R^2 values reaching up to 0.30.
- **Hotspot Analyses:**
 - **LST Z-score:** High-heat hotspots were concentrated in southern Brooklyn, eastern Bronx, and western Staten Island.
 - **Tree Canopy (Getis-Ord G_i^*):** Cold spots (low canopy) were detected in central Brooklyn and eastern Queens, while hot spots (dense canopy) were observed in northern Bronx and Staten Island.

9. CONCLUSION AND POLICY RECOMMENDATIONS

9.1 Conclusion

This project assessed the spatial distribution of urban tree canopy (UTC) in New York City for the year 2021 and evaluated its relationship with land surface temperature (LST) using remote sensing and spatial regression techniques. The findings demonstrate that while vegetation plays a clear role in urban cooling, its distribution is highly uneven across NYC's boroughs. Staten Island and the Bronx benefit from greater canopy cover, while central Brooklyn, lower Manhattan, and parts of Queens remain underserved.

The use of NDVI, LST, regression analysis, and spatial models (SLM, SEM, and GWR) provided both global and local insights into the tree canopy–temperature relationship. Though global models showed limited explanatory power, GWR revealed strong spatial heterogeneity—highlighting borough-specific variations and confirming the role of local land use and socioeconomic factors in influencing vegetation impact.

Hotspot analysis further emphasized the disparity in canopy coverage, with vulnerable areas suffering from both low tree presence and elevated LST. These results underscore the urgency of implementing data-driven urban forestry policies that target environmental equity.

9.2 Policy Recommendations

1. Prioritize Greening in Underserved Areas

Expand tree planting initiatives in canopy-deficient zones such as central Brooklyn and southern Queens, where UHI effects are most severe.

2. Adopt Equity-Based Urban Forestry Plans

Use spatial regression outputs to guide equitable tree canopy investment strategies, focusing on high-density, low-income neighborhoods with elevated heat exposure.

3. Integrate UTC Planning with Climate Resilience Policies

Incorporate tree canopy expansion into broader urban climate adaptation frameworks, including stormwater management, air quality enhancement, and energy efficiency.

4. Leverage Geospatial Tools for Monitoring and Assessment

Continue using GIS, remote sensing, and spatial models to track canopy growth, LST trends, and the effectiveness of greening programs over time.

5. Promoting Community Engagement and Stewardship

Involve local residents and stakeholders in planning, planting, and maintaining trees to ensure long-term success and inclusivity.

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